

Marked Poisson arrivals

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Let N be the number of customers arriving at a service point during a specified period, where $N \sim \text{Po}(\lambda)$. Each customer is served independently with probability $p \in (0, 1)$. The total number of customers served, S , is given by the random sum:

$$S = X_1 + \cdots + X_N,$$

where N and the sequence $\{X_i\}_{i \geq 1}$ are independent random variables, with $X_i \sim \text{Be}(p)$ for all $i \geq 1$. In this framework, each arrival is independently *marked* with probability p , and S is the total number of marked arrivals.

Previous results show that $S \sim \text{Po}(\lambda p)$ and $R = N - S \sim \text{Po}(\lambda q)$, where R is the number of non-marked arrivals. This note will confirm these results and demonstrate that S and R are independent, using characteristic functions.

Firstly, let us consider the joint characteristic function $M_{SR}(\omega_1, \omega_2)$. We have

$$M_{SR}(\omega_1, \omega_2) = \mathbb{E} \left(e^{i(\omega_1 S + \omega_2 R)} \right) = \mathbb{E} \left(e^{i(\omega_1 S + \omega_2 (N - S))} \right) = \mathbb{E} \left(e^{i((\omega_1 - \omega_2)S + \omega_2 N)} \right). \quad (1)$$

We calculate the expectation in (1) by first conditioning on N , as follows:

$$M_{SR}(\omega_1, \omega_2) = \mathbb{E} \left(\mathbb{E} \left(e^{i((\omega_1 - \omega_2)S + \omega_2 N)} \mid N \right) \right). \quad (2)$$

If $N = n$, then

$$\mathbb{E} \left(e^{i((\omega_1 - \omega_2)S + \omega_2 N)} \mid N = n \right) = \mathbb{E} \left(e^{i((\omega_1 - \omega_2)S + \omega_2 n)} \mid N = n \right) = e^{i\omega_2 n} \mathbb{E} \left(e^{i(\omega_1 - \omega_2)S} \mid N = n \right).$$

Moreover, we know that if $N = n$, the random variable $S = X_1 + \cdots + X_n$ is $\text{Bin}(n, p)$ -distributed. Therefore $\mathbb{E} \left(e^{i\omega S} \mid N = n \right) = (q + pe^{i\omega})^n$, because this conditional expectation is the characteristic function of that binomial distribution. Thus, we deduce that

$$\mathbb{E} \left(e^{i(\omega_1 - \omega_2)S} \mid N = n \right) = \left(q + pe^{i\omega} \right)^n \Big|_{\omega = \omega_1 - \omega_2} = \left(q + pe^{i(\omega_1 - \omega_2)} \right)^n.$$

Consequently,

$$\mathbb{E} \left(e^{i((\omega_1 - \omega_2)S + \omega_2 N)} \mid N = n \right) = e^{i\omega_2 n} \left(q + pe^{i(\omega_1 - \omega_2)} \right)^n = \left(pe^{i\omega_1} + qe^{i\omega_2} \right)^n.$$

In other words,

$$\mathbb{E} \left(e^{i((\omega_1 - \omega_2)S + \omega_2 N)} \mid N \right) = \left(pe^{i\omega_1} + qe^{i\omega_2} \right)^N. \quad (3)$$

Substituting (3) into (2) we obtain

$$M_{SR}(\omega_1, \omega_2) = \mathbb{E} \left(\left(pe^{i\omega_1} + qe^{i\omega_2} \right)^N \right) = G_N \left(pe^{i\omega_1} + qe^{i\omega_2} \right),$$

where $G_N(z) = \exp(\lambda(z - 1))$ is the probability generating function of the variable N , which is valid for all $z \in \mathbb{C}$. So we conclude that

$$M_{SR}(\omega_1, \omega_2) = \exp(\lambda(pe^{i\omega_1} + qe^{i\omega_2} - 1)).$$

Now, the marginal characteristic functions of S and R can be obtained as

$$M_S(\omega) = M_{SR}(\omega, 0) = \exp(\lambda(pe^{i\omega} + q - 1)) = \exp(\lambda p(e^{i\omega} - 1))$$

and

$$M_R(\omega) = M_{SR}(0, \omega) = \exp(\lambda(p + qe^{i\omega} - 1)) = \exp(\lambda q(e^{i\omega} - 1)),$$

respectively. Thus, we confirm that $S \sim \text{Po}(\lambda p)$ and $R \sim \text{Po}(\lambda q)$.

Furthermore,

$$\begin{aligned} M_S(\omega_1)M_S(\omega_2) &= \exp(\lambda p(e^{i\omega_1} - 1)) \cdot \exp(\lambda q(e^{i\omega_2} - 1)) \\ &= \exp(\lambda p e^{i\omega_1} + \lambda q e^{i\omega_2} - \lambda p - \lambda q) \\ &= \exp(\lambda(pe^{i\omega_1} + qe^{i\omega_2} - 1)) \\ &= M_{SR}(\omega_1, \omega_2), \end{aligned}$$

This factorisation demonstrates that S and R are independent random variables.